

ML notes6

Unsupervised learning:

Unsupervised learning is a type of machine learning in which a model learns patterns, structures, or relationships from data without being given labeled answers or target outcomes.

Clustering:

Clustering is the process of dividing a dataset into groups (clusters) based on similarity or distance measures without using predefined labels.

K-means clustering:

It's a method to divide a dataset into K clusters, where each data point belongs to the cluster with the nearest center called a centroid. An optimum value of K is obtained through elbow-method.

Elbow method:

The Elbow Method is a graphical technique for selecting the optimal number of clusters in K-Means by identifying the point at which increasing the number of clusters yields diminishing reductions in WCSS (Within-Cluster Sum of Squares).

Hierarchical Clustering:

Hierarchical clustering is an unsupervised clustering technique that builds a hierarchy of clusters by either repeatedly merging smaller clusters (**agglomerative**) or repeatedly splitting larger clusters (**divisive**), often visualized using a dendrogram.

Unlike **K-Means**, it does not require specifying the number of clusters beforehand.

A **dendrogram** is a tree diagram that shows how clusters are merged or split.

Agglomerative Clustering (Bottom-Up Approach):

- Starts with each data point as its own cluster.
- Repeatedly merges the most similar clusters.
- Continues until all points belong to a single cluster or a stopping criterion is met.

Divisive Clustering (Top-Down Approach):

- Starts with all data points in one cluster.
- Repeatedly splits clusters into smaller clusters.
- Continues until each point is in its own cluster or the desired number of clusters is reached.

DBSCAN (Density Based Spatial Clustering of Applications with Noise):

DBSCAN is a clustering algorithm that groups together points that are closely packed and marks isolated points as outliers (noise).

i) Epsilon (ϵ):

The maximum distance within which points are considered neighbors.

ii) MinPts (Minimum Points):

The minimum number of points required within the ϵ -neighborhood to form a dense region.

iii) Core Point:

A point is a core point if it has at least MinPts points (including itself) within its ϵ -neighborhood.

Ex:

If $\epsilon = 2$ and MinPts = 5, a point having 5 or more points within distance 2 is a core point.

iv) Border Point:

Has fewer than MinPts neighbors but is reachable from a core point.

v) Noise Point:

Neither a core point nor reachable from any core point.

“When we have a dataset which has too much of volume of data or a lot of density of the data, with lot of datapoints at one place, we can use DBSCAN else we can use K-means or Hierarchical clusters.”

Evaluation of Clusters:

1. External Evaluation

- Measures how well the clusters match the true labels or known classes.
- Requires ground-truth data.

Ex: Comparing clustered students with their actual departments.

2. Internal Evaluation

- Measures the quality of clusters using only the data itself, without true labels.
- Checks whether points within a cluster are similar and clusters are well separated.

Ex: Evaluating how compact and distinct the clusters are.

3. Reliability Evaluation

- Measures how consistent the clustering results are when the algorithm is run multiple times or the data changes slightly.

Ex: Getting similar clusters in repeated runs of K-Means.

Natural Language Processing (NLP):

Natural Language Processing (NLP) is a branch of artificial intelligence that helps computers understand, interpret, and generate human language.

1. Tokenization

- The process of breaking text into smaller units called tokens (words, sentences or phrases).

Example:

"Machine learning is fun" → ["Machine", "learning", "is", "fun"].

2. Part-of-Speech (POS) Tagging

- Identifying the grammatical role of each word in a sentence (noun, verb, adjective, etc.).

Example:

"She runs fast"

- She → Pronoun

- runs → Verb
- fast → Adverb

3. Stop Words

- Common words that are often removed because they carry little meaning.

Examples:

"the", "is", "and", "in", "of"

4. Stemming

- Reduces a word to its root form by removing prefixes or suffixes, sometimes producing non-dictionary words.

Example:

- Playing → Play
- Studies → Studi

5. Lemmatization

- Reduces a word to its dictionary (base) form while considering its meaning and grammar.

Example:

- Running → Run
- Better → Good

6. Named Entity Recognition (NER):

- Identifies and classifies named entities such as people, organizations, locations, dates, etc., in text.

Example:

"Apple released a new iPhone."

- Apple - Organization
- iPhone - Product

NER helps distinguish meanings. For example:

- "Apple is a fruit." - Apple = Fruit (not an organization)

- "Apple launched a phone." - Apple = Company

BIO Tagging in NER

BIO tagging labels words based on their position in an entity.

- **B (Beginning):** First word of an entity.
- **I (Inside):** Continuation of the same entity.
- **O (Outside):** Not part of any entity.

Bag of Words (Bow):

- A method that represents text by counting how many times each word appears, ignoring grammar and order.

TF-IDF (Term Frequency - Inverse Document Frequency):

- A method that gives importance to words based on how often they appear in a document and how rare they are across all documents.

N-grams:

- A way of representing text using sequences of N words together.
- **Bigram (2-gram):** 2 words together
Example: "machine learning"
- **Trigram (3-gram):** 3 words together
Example: "natural language processing"
- **Skip-gram:** Words that are related but not necessarily next to each other.

CBOW (Continuous Bag of Words):

- A model that predicts a target word using surrounding words (context).

Ex:

"The cat ____ on the mat" → predicts "sat"

Word2Vec:

- A technique that converts words into numerical vectors (embeddings) so that similar words have similar meanings in vector space.

Ex:

"king" and "queen" will be closer in meaning than "king" and "car".

RNN: Recursive neural network:

It's used for text data. It remembers the context in the output.

Issue: As the words increase the RNN also increases eventually it becomes complex.

LSTM:

Input gate, output gate, forget gate

Only remembers the data in cell state. More context awareness.

GRU:

Reset gate, Output gate. Long term memory.